

Proposed Prescribing Information
For the use of a Registered Medical Practitioner or a Hospital or a Laboratory

GABATOR CAPSULES 300mg/ GABATOR CAPSULES 400mg
GABAPENTIN CAPSULES 300 mg / GABAPENTIN CAPSULES 400 mg

BRAND OR PRODUCT NAME

GABATOR CAPSULES 300mg/ GABATOR CAPSULES 400mg

NAME AND STRENGTH OF ACTIVE SUBSTANCE(S)

GABATOR CAPSULES 300mg

Each hard Gelatin Capsule contains

Gabapentin USP.....300 mg

Excipients.....q.s

Approved colours used in the capsule shell

GABATOR CAPSULES 400mg

Each hard Gelatin Capsule contains

Gabapentin USP400 mg

Excipients.....q.s

Approved colours used in the capsule shell

PRODUCT DESCRIPTION

GABATOR CAPSULES 300mg:

Yellow/Yellow coloured, size '1', capsules printed with '300' in black colour on the cap and Torrent logo (Square emblem only) in black colour on the body of capsule shell containing white to off-white powder

GABATOR CAPSULES 400mg:

Orange/Orange coloured, size '0', capsules printed with '400' in black colour on the cap and Torrent logo (Square emblem only) in black colour on the body of capsule shell containing white to offwhite powder

DOSAGE FORM

Hard Gelatin Capsule

PHARMACODYNAMICS / PHARMACOKINETICS

Pharmacodynamics

Pharmacotherapeutic groups: Other antiepileptics ATC code: N03AX12

Mechanism of action

Gabapentin readily enters the brain and prevents seizures in a number of animal models of epilepsy.

Gabapentin does not possess affinity for either GABAA or GABAB receptor nor does it alter the metabolism of GABA. It does not bind to other neurotransmitter receptors of the brain and does not interact with sodium channels. Gabapentin binds with high affinity to the $\alpha 2\delta$ (alpha-2-delta) subunit of voltage-gated calcium channels and it is proposed that binding to the $\alpha 2\delta$ subunit may be involved in gabapentin's anti-seizure effects in animals. Broad panel screening does not suggest any other drug targets other than $\alpha 2\delta$.

Evidence from several pre-clinical models inform that the pharmacological activity of gabapentin may be mediated via binding to $\alpha 2\delta$ through a reduction in release of excitatory neurotransmitters in regions of the central nervous system. Such activity may underlie gabapentin's anti-seizure activity. The relevance of these actions of gabapentin to the anticonvulsant effects in humans remains to be established.

Gabapentin also displays efficacy in several pre-clinical animal pain models. Specific binding of gabapentin to the $\alpha 2\delta$ subunit is proposed to result in several different actions that may be responsible for analgesic activity in animal models. The analgesic activities of gabapentin may occur in the spinal cord as well as at higher brain centers through interactions with descending pain inhibitory pathways. The relevance of these pre-clinical properties to clinical action in humans is unknown.

Pharmacokinetics

Absorption

Following oral administration, peak plasma gabapentin concentrations are observed within 2 to 3 hours. Gabapentin bioavailability (fraction of dose absorbed) tends to decrease with increasing dose. Absolute bioavailability of a 300 mg capsule is approximately 60%. Food, including a high-fat diet, has no clinically significant effect on gabapentin pharmacokinetics.

Gabapentin pharmacokinetics are not affected by repeated administration.

Distribution

Gabapentin is not bound to plasma proteins and has a volume of distribution equal to 57.7 litres. In patients with epilepsy, gabapentin concentrations in cerebrospinal fluid (CSF) are approximately 20% of corresponding steady-state trough plasma concentrations. Gabapentin is present in the breast milk of breast-feeding women.

Biotransformation

There is no evidence of gabapentin metabolism in humans. Gabapentin does not induce hepatic mixed function oxidase enzymes responsible for drug metabolism.

Elimination

Gabapentin is eliminated unchanged solely by renal excretion. The elimination half-life of gabapentin is independent of dose and averages 5 to 7 hours.

In elderly patients, and in patients with impaired renal function, gabapentin plasma clearance is reduced. Gabapentin elimination-rate constant, plasma clearance, and renal clearance are directly proportional to creatinine clearance.

Gabapentin is removed from plasma by haemodialysis. Dosage adjustment in patients with compromised renal function or undergoing haemodialysis is recommended.

In general, plasma gabapentin concentrations in children > 5 years of age are similar to those in adults when dosed on a mg/kg basis.

In patients aged between 1 month and 48 months, an approximately 30% lower exposure (AUC), lower C_{max} and higher clearance per body weight have been observed in comparison to children older than 5 years.

Linearity/non-linearity

Gabapentin bioavailability (fraction of dose absorbed) decreases with increasing dose which imparts non-linearity to pharmacokinetic parameters which include the bioavailability parameter (F) e.g. A_e%, CL/F, V_d/F. Elimination pharmacokinetics (pharmacokinetic parameters which do not include F such as CL_r and T_{1/2}), are best described by linear pharmacokinetics. Steady state plasma gabapentin concentrations are predictable from single-dose data.

INDICATION

Epilepsy

Gabapentin is indicated as adjunctive therapy in the treatment of partial seizures with and without secondary generalization in adults and children age 3 years. Safety and effectiveness for adjunctive therapy in pediatric patients below the age of 3 years have not been established.

Treatment of peripheral neuropathic pain

Gabapentin is indicated for the treatment of peripheral neuropathic pain which includes diabetic neuropathy, post-herpetic neuralgia and trigeminal neuralgia in adult's age 18 years and above. Safety and effectiveness in patients below the age of 18 years have not been established.

RECOMMENDED DOSAGE AND MODE OF ADMINISTRATION

General

Gabapentin is given orally with or without food.

When in the judgment of the clinician there is a need for dose reduction, discontinuation, or substitution with an alternative medication, this should be done gradually over a minimum of one week.

Epilepsy

Adults and pediatric patients over 12 years of age

The effective dosing range was 900 to 3600 mg/day. Therapy may be initiated by administering 300 mg three times a day (TID) on Day 1, or by titrating the dose as described in Table 1. Thereafter, the dose can be increased in three equally divided doses up to a maximum dose of 3600 mg/day. The maximum time between doses in the three times a day (TID) schedule should not exceed 12 hours to prevent breakthrough convulsions.

TABLE 1			
DOSING CHART- INITIAL TITRATION			
Dose	Day 1	Day 2	Day 3
900 mg	300 mg QD ^a	300 mg BID ^b	300 mg TID ^c

^aQD = once a day ^bBID

= two times a day

^cTID = three times a day

Pediatric Patients Age 3–12 years

The starting dose should range from 10 to 15 mg/kg/day given in equally divided doses (three times a day), and the effective dose reached by upward titration over a period of approximately three days. The effective dose of gabapentin in pediatric patients age 5 years and older is 25 to 35 mg/kg/day given in equally divided doses (three times a day). The effective dose in pediatric patients ages 3 to less than 5 years is 40 mg/kg/day given in equally divided doses (three times a day). Dosages up to 50 mg/kg/day have been well tolerated in a long-term clinical study. The maximum time interval between doses should not exceed 12 hours.

It is not necessary to monitor gabapentin plasma concentrations to optimize gabapentin therapy. Further, gabapentin may be used in combination with other antiepileptic drugs without concern for alteration of the plasma concentrations of gabapentin or serum concentrations of other antiepileptic drugs.

Neuropathic pain in adults

The starting dose is 900 mg/day given as three equally divided doses, and increased if necessary, based on response, up to a maximum dose of 3600 mg/day. Therapy should be initiated by titrating the dose as described in TABLE 1.

Dosage adjustment in impaired renal function for patients with neuropathic pain or epilepsy

Dosage adjustment is recommended in patients with compromised renal function as described in TABLE 2 and/or those undergoing hemodialysis.

TABLE 2: DOSAGE OF GABAPENTIN IN ADULTS BASED ON RENAL FUNCTION	
Creatinine Clearance (mL/min)	Total Daily Dose ^a (mg/day)
≥80	900-3600
50-79	600-1800
30-49	300-900
15-29	150 ^b -600
<15	150 ^b -300

^a Total daily dose should be administered as a TID regimen. Doses used to treat patients with normal renal function (creatinine clearance >80 mL/min) range from 900 to 3600 mg/day. Reduced dosages are for patients with renal impairment (creatinine clearance < 79 mL/min). ^b To be administered as 300mg every other day.

Dosage adjustment in patients undergoing hemodialysis

For patients undergoing hemodialysis who have never received gabapentin, a loading dose of 300 to 400mg is recommended, then 300 mg of gabapentin following each 4 hours of hemodialysis.

Mode of Administration

Oral

CONTRAINDICATIONS

Hypersensitivity to gabapentin or any component of the product.

WARNINGS AND PRECAUTIONS

Drug Rash with Eosinophilia and Systemic Symptoms (DRESS)

Severe, life-threatening, systemic hypersensitivity reactions such as Drug rash with eosinophilia and systemic symptoms (DRESS) have been reported in patients taking antiepileptic drugs including gabapentin.

It is important to note that early manifestations of hypersensitivity, such as fever or lymphadenopathy, may be present even though rash is not evident. If such signs or symptoms are present, the patient should be evaluated immediately. Gabapentin should be discontinued if an alternative etiology for the signs or symptoms cannot be established.

Anaphylaxis

Gabapentin can cause anaphylaxis. Signs and symptoms in reported cases have included difficulty breathing, swelling of the lips, throat, and tongue, and hypotension requiring emergency treatment. Patients should be instructed to discontinue gabapentin and seek immediate medical care should they experience signs or symptoms of anaphylaxis.

Suicidal ideation and behaviour

Suicidal ideation and behaviour have been reported in patients treated with anti-epileptic agents in several indications. The mechanism of this risk is not known and the available data do not exclude the possibility of an increased risk for gabapentin.

Therefore patients should be monitored for signs of suicidal ideation and behaviours and appropriate treatment should be considered. Patients (and caregivers of patients) should be advised to seek medical advice should signs of suicidal ideation or behaviour emerge.

Acute pancreatitis

If a patient develops acute pancreatitis under treatment with gabapentin, discontinuation of gabapentin should be considered.

Seizures

Although there is no evidence of rebound seizures with gabapentin, abrupt withdrawal of anticonvulsants in epileptic patients may precipitate status epilepticus.

As with other antiepileptic medicinal products, some patients may experience an increase in seizure frequency or the onset of new types of seizures with gabapentin.

As with other anti-epileptics, attempts to withdraw concomitant anti-epileptics in treatment refractive patients on more than one antiepileptic, in order to reach gabapentin monotherapy, have a low success rate.

Gabapentin is not considered effective against primary generalized seizures such as absences and may aggravate these seizures in some patients. Therefore, gabapentin should be used with caution in patients with mixed seizures including absences.

Gabapentin treatment has been associated with dizziness and somnolence, which could increase the occurrence of accidental injury (fall). Patients should be advised to exercise caution until they are familiar with the potential effects of the medication

Concomitant use with opioids

Patients who require concomitant treatment with opioids should be carefully observed for signs of central nervous system (CNS) depression, such as somnolence, sedation and respiratory depression. Patients who use gabapentin and morphine concomitantly may experience increases in gabapentin concentrations. The dose of gabapentin or opioids should be reduced appropriately.

Respiratory depression

Gabapentin has been associated with severe respiratory depression. Patients with compromised respiratory function, respiratory or neurological disease, renal impairment, concomitant use of CNS depressants and the elderly might be at higher risk of experiencing this severe adverse reaction. Dose adjustments might be necessary in these patients.

Paediatric population

The effects of long-term (greater than 36 weeks) gabapentin therapy on learning, intelligence, and development in children and adolescents have not been adequately studied. The benefits of prolonged therapy must therefore be weighed against the potential risks of such therapy.

Abuse and dependence

Carefully evaluate patients for a history of drug abuse and observe them for possible signs of gabapentin abuse e.g. drug-seeking behaviour, dose escalation, development of tolerance.

Laboratory tests

False positive readings may be obtained in the semi-quantitative determination of total urine protein by dipstick tests. It is therefore recommended to verify such a positive dipstick test result by methods based on a different analytical principle such as the Biuret method, turbidimetric or dyebinding methods, or to use these alternative methods from the beginning.

Effects on ability to drive and use machines

Gabapentin may have minor or moderate influence on the ability to drive and use machines. Gabapentin acts on the central nervous system and may cause drowsiness, dizziness or other related symptoms. Even, if they were only of mild or moderate degree, these undesirable effects could be potentially dangerous in patients driving or operating machinery. This is especially true at the beginning of the treatment and after increase in dose.

INTERACTIONS WITH OTHER MEDICAMENTS

There are spontaneous and literature case reports of respiratory depression and/or sedation associated with gabapentin and opioid use. In some of these reports, the authors considered this a particular concern with the combination of gabapentin and opioids, especially in elderly patients.

Therefore, patients who require concomitant treatment with opioids should be carefully observed for signs of CNS depression, such as somnolence, sedation and respiratory depression and the dose of gabapentin or opioid should be reduced appropriately.

No interaction between gabapentin and phenobarbital, phenytoin, valproic acid or carbamazepine has been observed.

Co-administration of gabapentin with oral contraceptives containing norethindrone and/or ethinyl estradiol, does not influence the steady-state pharmacokinetics of either component.

Co-administration of gabapentin with antacids containing aluminium and magnesium, reduces gabapentin bioavailability up to 24%. It is recommended that gabapentin be taken at the earliest two hours following antacid administration.

Renal excretion of gabapentin is unaltered by probenecid.

A slight decrease in renal excretion of gabapentin that is observed when it is co-administered with cimetidine is not expected to be of clinical importance.

STATEMENT ON USAGE DURING PREGNANCY AND LACTATION**Pregnancy:**

There are no adequate data and well controlled studies in pregnant women. Because animal reproduction studies are not always predictive of human response, this drug should be used during pregnancy only if the potential benefit to the patient justifies the potential risk to the fetus.

Lactation:

Gabapentin is excreted in human milk. Because the effect on the nursing infant is unknown, caution should be exercised when gabapentin is administered to a nursing mother. Gabapentin should be used in nursing mothers only if the benefits clearly outweigh the risks.

ADVERSE EFFECTS / UNDESIRABLE EFFECTS

System organ class	Adverse drug reactions
Infections and infestations	
Very Common	viral infection
Common	pneumonia, respiratory infection, urinary tract infection, infection, otitis media
Blood and the lymphatic system disorders	
Common	Leucopenia
Not known	<i>Thrombocytopenia</i>
Immune system disorders	
Uncommon	allergic reactions (e.g. urticaria)
Not known	<i>hypersensitivity syndrome(a systemic reaction with a variable presentation that can include fever, rash, hepatitis, lymphadenopathy, eosinophilia, and sometimes other signs and symptoms), anaphylaxis</i>
Metabolism and nutrition disorders	
Common	anorexia, increased appetite
Uncommon	hyperglycaemia (most often observed in patients with diabetes)
Rare	hypoglycaemia (most often observed in patients with diabetes)
Not known	<i>hyponatraemia</i>
Psychiatric disorders	
Common	hostility, confusion and emotional lability, depression, anxiety, nervousness, thinking abnormal
Uncommon	agitation
Not known	<i>Hallucinations</i>
Nervous system disorders	
Very Common	somnolence, dizziness, ataxia
Common	convulsions, hyperkinesias, dysarthria, amnesia, tremor, insomnia, headache, sensations such as paresthesia, hypaesthesia, coordination abnormal, nystagmus, increased, decreased, or absent reflexes

Uncommon	hypokinesia, mental impairment
Rare	loss of consciousness
Not known	<i>other movement disorders (e.g. choreoathetosis, dyskinesia, dystonia)</i>
Eye disorders	
Common	visual disturbances such as amblyopia, diplopia
Ear and labyrinth disorders	
Common	Vertigo
Not known	<i>Tinnitus</i>
Cardiac disorders	
Uncommon	Palpitations
Vascular disorders	
Common	hypertension, vasodilatation
Respiratory, thoracic and mediastinal disorders	
Common	dyspnoea, bronchitis, pharyngitis, cough, rhinitis
Rare	Respiratory depression
Gastrointestinal disorders	
Common	vomiting, nausea, dental abnormalities, gingivitis, diarrhoea, abdominal pain, dyspepsia, constipation, dry mouth or throat, flatulence
Not known	<i>Pancreatitis</i>
Hepatobiliary disorders	
Not known	<i>hepatitis, jaundice</i>
Skin and subcutaneous tissue disorders	
Common	facial oedema, purpura most often described as bruises resulting from physical trauma, rash, pruritus, acne
Not known	<i>Stevens-Johnson syndrome, angioedema, erythema multiforme, alopecia, drug rash with eosinophilia and systemic symptoms</i>
Musculoskeletal and connective tissue disorders	
Common	arthralgia, myalgia, back pain, twitching
Not known	<i>rhabdomyolysis, myoclonus</i>
Renal and urinary disorder	
Not known	acute renal failure, incontinence
Reproductive system and breast disorders	

Common	Impotence
Not known	<i>breast hypertrophy, gynaecomastia, sexual dysfunction (including changes in libido, ejaculation disorders and anorgasmia)</i>
General disorders and administration site conditions	
Very Common	fatigue, fever
Common	peripheral oedema, abnormal gait, asthenia, pain, malaise, flu syndrome
Uncommon	generalized oedema
Not known	<i>withdrawal reactions (mostly anxiety, insomnia, nausea, pains, sweating), chest pain. Sudden unexplained deaths have been reported where a causal relationship to treatment with gabapentin has not been established.</i>
Investigations	
Common	WBC (white blood cell count) decreased, weight gain
Uncommon	elevated liver function tests SGOT (AST), SGPT (ALT) and bilirubin
Not known	<i>blood creatine phosphokinase increased</i>
Injury, poisoning and procedural complications	
Common	accidental injury, fracture, abrasion
Uncommon	fall
Post-marketing experience: Dysphagia	

Under treatment with gabapentin cases of acute pancreatitis were reported. Causality with gabapentin is unclear.

In patients on haemodialysis due to end-stage renal failure, myopathy with elevated creatine kinase levels has been reported.

OVERDOSE AND TREATMENT

Acute, life-threatening toxicity has not been observed with gabapentin overdoses of up to 49 grams. Symptoms of the overdoses included dizziness, double vision, slurred speech, drowsiness, lethargy and mild diarrhea. All patients recovered fully with supportive care. Reduced absorption of gabapentin at higher doses may limit drug absorption at the time of overdosing and, hence, minimize toxicity from overdoses.

Although gabapentin can be removed by hemodialysis, based on prior experience it is usually not required. However, in patients with severe renal impairment, hemodialysis may be indicated.

An oral lethal dose of gabapentin was not identified in mice and rats given doses as high as 8000 mg/kg. Signs of acute toxicity in animals included ataxia, labored breathing, ptosis, hypoactivity, or excitation.

STORAGE CONDITIONS

Store below 30°C. Protected from Moisture.

PACKAGING AVAILABLE

Gabator capsules are packed in Alu-Alu blister of 10 capsules. Such blisters containing 10 capsules are packed into boxes of 30's and 100's.

DATE OF REVISION OF PACKAGE INSERT

27-01-2021

NAME AND ADDRESS OF MANUFACTURER

TORRENT PHARMACEUTICALS LIMITED

INDRAD-382 721, DIST. MEHSANA, INDIA.